

Practice Sheet: ROUND, ROUNDUP, and ROUNDDOWN Functions

This practice sheet helps you master the **ROUND**, **ROUNDUP**, and **ROUNDDOWN** functions in Excel.

These functions are used to control decimal places and round numbers in billing, salary, accounting, stock, and office calculation work.

This module uses an invoice billing dataset for practical office-style exercises using rounding functions.

New Functions: ROUND, ROUNDUP, and ROUNDDOWN

=ROUND(number, num_digits)

The **ROUND** function rounds a number to a specified number of digits.

=ROUNDUP(number, num_digits)

The **ROUNDUP** function always rounds a number upward, away from zero.

=ROUNDDOWN(number, num_digits)

The **ROUNDDOWN** function always rounds a number downward, towards zero.

Examples:

=ROUND(245.678,2)

This returns **245.68**.

=ROUNDUP(245.12,0)

This returns **246**.

=ROUNDDOWN(245.98,0)

This returns **245**.

Step 1: Set Up the Main Data Table

In **Sheet1**, starting from cell **A1**, enter the following table.

Invoice No.	Customer Name	Product Name	Category	Qty	Unit Price (₹)	Dis-count %	GST %
INV001	Ankit Sharma	Wireless Mouse	Computer	2	625.75	5%	18%
INV002	Rima Das	A4 Notebook Set	Stationery	5	145.40	3%	12%
INV003	Tanmay Bora	USB Keyboard	Computer	1	875.90	4%	18%
INV004	Priyanka Dutta	Formal Shirt	Clothing	3	799.55	8%	12%
INV005	Arindam Saikia	LED Desk Lamp	Electronics	2	1245.80	6%	18%
INV006	Moumita Roy	Ball Pen Box	Stationery	4	235.65	2%	12%
INV007	Bijoy Kalita	Office Chair	Furniture	1	4250.75	10%	18%
INV008	Nisha Sharma	Printer Ink	Computer	2	1125.30	5%	18%
INV009	Imran Ali	Cotton Kurta	Clothing	2	945.25	7%	12%

Invoice No.	Customer Name	Product Name	Category	Qty	Unit Price (₹)	Dis-count %	GST %
INV010	Rakesh Nath	Bluetooth Speaker	Electronics	1	2140.60	6%	18%
INV011	Puja Das	Stapler Set	Stationery	6	85.75	2%	12%
INV012	Akash Baruah	Wooden Table	Furniture	1	6850.90	12%	18%
INV013	Manisha Paul	Ladies Handbag	Accessories	2	1320.45	8%	12%
INV014	Rohit Singh	Power Bank	Electronics	2	1599.99	5%	18%
INV015	Sangeeta Gogoi	File Folder Pack	Stationery	8	72.60	1%	12%
INV016	Debajit Deka	Laptop Stand	Computer	1	975.85	4%	18%
INV017	Kavita Roy	Plastic Storage Box	Household	3	399.50	5%	18%
INV018	Suman Chetia	Denim Jeans	Clothing	2	1395.75	9%	12%
INV019	Farhana Begum	Smart Watch	Electronics	1	3499.45	7%	18%
INV020	Kunal Sharma	Rice Cooker	Household	1	2899.90	6%	18%
INV021	Deepika Kalita	Marker Pen Set	Stationery	5	168.25	3%	12%
INV022	Rajib Das	Sofa Cushion	Furniture	4	545.60	8%	18%
INV023	Neha Kapoor	External Hard Disk	Computer	1	4899.75	5%	18%
INV024	Hiren Bora	Kitchen Knife Set	Household	2	675.45	4%	18%
INV025	Alisha Ahmed	Woollen Scarf	Clothing	3	425.35	6%	12%
INV026	Pranjal Saikia	Ceiling Fan	Electronics	1	2199.80	7%	18%
INV027	Karishma Dey	Geometry Box	Stationery	10	95.25	2%	12%
INV028	Vikram Sinha	Study Table	Furniture	1	5250.55	10%	18%
INV029	Meera Nair	Earphones	Electronics	2	745.95	5%	18%
INV030	Abhishek Roy	T-Shirt Pack	Clothing	4	599.49	8%	12%
INV031	Sneha Dutta	Calculator	Stationery	3	325.80	3%	12%
INV032	Rahul Choudhury	Monitor 24 inch	Computer	1	9850.75	6%	18%
INV033	Ankita Mahanta	Dinner Plate Set	Household	2	899.65	5%	18%
INV034	Nilotpall Das	Bookshelf	Furniture	1	3650.40	9%	18%
INV035	Jyoti Devi	Cotton Saree	Clothing	1	1750.90	7%	12%

Invoice No.	Customer Name	Product Name	Category	Qty	Unit Price (₹)	Dis-count %	GST %
INV036	Manash Sarma	WiFi Router	Computer	1	2450.35	4%	18%
INV037	Rituparna Bora	Water Bottle	Household	6	185.75	3%	18%
INV038	Wasim Ahmed	Desk Organizer	Stationery	4	249.99	2%	12%
INV039	Pankaj Dutta	Mobile Charger	Electronics	3	499.50	5%	18%
INV040	Sonali Gogoi	Sports Shoes	Clothing	1	2245.85	8%	12%
INV041	Chandan Roy	Dining Chair	Furniture	2	1850.25	10%	18%
INV042	Bipasha Das	Mixer Grinder	Household	1	3299.99	6%	18%
INV043	Nirmal Kalita	Tablet Cover	Accessories	2	445.70	4%	12%
INV044	Aditi Sharma	Office Bag	Accessories	1	1540.35	7%	12%
INV045	Rajashree Deka	Pencil Box	Stationery	7	115.60	2%	12%
INV046	Partha Pratim	Laptop Backpack	Accessories	1	1899.80	8%	12%
INV047	Mitali Saikia	Wall Clock	Household	2	695.45	5%	18%
INV048	Saurav Bora	Wireless Keyboard	Computer	1	1455.95	4%	18%
INV049	Monika Paul	Bedsheet Set	Household	2	1299.40	6%	18%
INV050	Diganta Gogoi	Extension Board	Electronics	3	375.85	3%	18%

Step 2: Create Calculation Columns

In **Sheet1**, add the following headings:

Cell Heading to Enter

I1	Gross Amount
J1	Discount Amount
K1	Net Before GST
L1	GST Amount
M1	Final Bill Rounded
N1	Round Up Bill
O1	Round Down Bill
P1	Difference

Step 3: Calculate Gross Amount

In cell **I2**, enter the following formula:

=E2*F2

Drag the formula down to cell **I51**.

This calculates the gross amount before discount.

Step 4: Calculate Discount Amount Using ROUND

In cell **J2**, enter the following formula:

=ROUND(I2*G2,2)

Drag the formula down to cell **J51**.

This calculates the discount amount rounded to **2 decimal places**.

Step 5: Calculate Net Amount Before GST

In cell **K2**, enter the following formula:

=I2-J2

Drag the formula down to cell **K51**.

This calculates the amount after discount but before GST.

Step 6: Calculate GST Amount Using ROUND

In cell **L2**, enter the following formula:

=ROUND(K2*H2,2)

Drag the formula down to cell **L51**.

This calculates GST amount rounded to **2 decimal places**.

Step 7: Calculate Final Bill Using ROUND

In cell **M2**, enter the following formula:

=ROUND(K2+L2,0)

Drag the formula down to cell **M51**.

This calculates the final bill rounded to the nearest whole rupee.

Step 8: Calculate Round Up Bill Using ROUNDUP

In cell **N2**, enter the following formula:

=ROUNDUP(K2+L2,0)

Drag the formula down to cell **N51**.

This calculates the final bill by always rounding up to the next whole rupee.

Step 9: Calculate Round Down Bill Using ROUNDDOWN

In cell **O2**, enter the following formula:

=ROUNDDOWN(K2+L2,0)

Drag the formula down to cell **O51**.

This calculates the final bill by always rounding down to the lower whole rupee.

Step 10: Calculate Difference Between ROUNDUP and ROUNDDOWN

In cell **P2**, enter the following formula:

=N2-02

Drag the formula down to cell **P51**.

This shows the difference between the rounded up bill and the rounded down bill.

Step 11: Create Sheet2 for Rounding Practice

Create a new sheet named **Rounding Practice**.

All the following practice formulas should be entered in the **Rounding Practice** sheet.

Step 12: Round Unit Prices to Whole Rupees

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
A1	Unit Prices Rounded
A2	=ROUND(Sheet1!F2:F51,0)

This rounds all unit prices to the nearest whole rupee.

Step 13: Round Unit Prices to One Decimal Place

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
D1	Unit Prices One Decimal
D2	=ROUND(Sheet1!F2:F51,1)

This rounds all unit prices to **1 decimal place**.

Step 14: Round Discount Amounts to Whole Rupees

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
G1	Discount Rounded
G2	=ROUND(Sheet1!J2:J51,0)

This rounds discount amounts to the nearest whole rupee.

Step 15: Round GST Amounts to Whole Rupees

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
J1	GST Rounded
J2	=ROUND(Sheet1!L2:L51,0)

This rounds GST amounts to the nearest whole rupee.

Step 16: Round Up Unit Prices

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
A60	Unit Prices Round Up
A61	=ROUNDUP(Sheet1!F2:F51,0)

This rounds every unit price upward to the next whole rupee.

Step 17: Round Down Unit Prices

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
D60	Unit Prices Round Down
D61	=ROUNDDOWN(Sheet1!F2:F51,0)

This rounds every unit price downward to the lower whole rupee.

Step 18: Round Final Bills to Nearest Ten

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
G60	Final Bills Nearest Ten
G61	=ROUND(Sheet1!M2:M51,-1)

This rounds final bills to the nearest **10 rupees**.

Step 19: Round Up Final Bills to Nearest Ten

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
J60	Final Bills Round Up to Ten
J61	=ROUNDUP(Sheet1!M2:M51,-1)

This rounds final bills upward to the next multiple of **10**.

Step 20: Round Down Final Bills to Nearest Ten

In the **Rounding Practice** sheet, enter the following:

Cell	Value to Enter
A120	Final Bills Round Down to Ten
A121	=ROUNDDOWN(Sheet1!M2:M51,-1)

This rounds final bills downward to the lower multiple of **10**.

Practice Questions

Use the **ROUND**, **ROUNDUP**, and **ROUNDDOWN** functions to solve the following questions. Use the exact cells given below in the **Rounding Practice** sheet so that everyone's workbook remains uniform.

Cell	Task
A180	Round all Unit Prices to the nearest whole rupee.
A240	Round all Unit Prices to 1 decimal place .
A300	Round all Gross Amounts to the nearest whole rupee.
A360	Round all Discount Amounts to the nearest whole rupee.
A420	Round all GST Amounts to the nearest whole rupee.
A480	Round all Net Before GST values to the nearest whole rupee.
A540	Round all Final Bills to the nearest whole rupee.
A600	Round up all Unit Prices to the next whole rupee.
A660	Round down all Unit Prices to the lower whole rupee.
A720	Round up all Final Bills to the next whole rupee.
A780	Round down all Final Bills to the lower whole rupee.
A840	Round all Final Bills to the nearest 10 .
A900	Round up all Final Bills to the next multiple of 10 .
A960	Round down all Final Bills to the lower multiple of 10 .
A1020	Round all Gross Amounts to the nearest 100 .
A1080	Round up all Gross Amounts to the next multiple of 100 .
A1140	Round down all Gross Amounts to the lower multiple of 100 .
A1200	Round all GST Amounts to 1 decimal place .
A1260	Round all Discount Amounts to 1 decimal place .
A1320	Calculate the difference between ROUNDUP Final Bill and ROUNDDOWN Final Bill .

Extra Challenge

Create a new sheet named **Dynamic Rounding**.

In the **Dynamic Rounding** sheet, create the following criteria table in cells **A1:B2**:

Cell	Value to Enter
A1	Criteria
B1	Value
A2	Decimal Places
B2	0

In cell **A5**, type:

Dynamic ROUND Result

In cell **A6**, enter the following formula:

=ROUND(Sheet1!M2:M51,B2)

Now change the value in cell **B2** to **1**, **0**, or **-1** and observe how the rounded result changes.

Homework

Create a new sheet named **Homework**. In the Homework sheet, create a new table of **50 salary records** starting from cell **A1** with the following columns:

Emp ID	Name	Department	Basic Salary	Allowance	Deduction	Bonus %
--------	------	------------	--------------	-----------	-----------	---------

Then use **ROUND**, **ROUNDUP**, and **ROUNDDOWN** functions to complete the following tasks. Use the exact cells given below in the **Homework** sheet.

Cell	Task
I1	Calculate Gross Salary.
J1	Calculate Bonus Amount using ROUND to 2 decimal places.
K1	Calculate Net Salary.
L1	Round Net Salary to the nearest whole rupee.
M1	Round up Net Salary to the next whole rupee.
N1	Round down Net Salary to the lower whole rupee.
I60	Round Net Salary to the nearest 10.
I120	Round up Net Salary to the next multiple of 10.
I180	Round down Net Salary to the lower multiple of 10.
I240	Round Gross Salary to the nearest 100.
I300	Round up Gross Salary to the next multiple of 100.
I360	Round down Gross Salary to the lower multiple of 100.

Save your workbook as **Excel_Practice_Module17.xlsx** and submit it to your instructor.

End of Module 17